

4.7 Specialist techniques, tools, equipment and processes that can be used to shape, fabricate, construct and assemble a high-quality thermoforming and thermosetting polymers prototype	Application, advantages and disadvantages, of the following specialist techniques when manufacturing products, in order to be able to discriminate between them and select appropriately for use.
	4.7.1 Tools and equipment: - a hand tools - b machinery - c digital design and manufacture.
	4.7.2 Shaping: - a laser cutting and engraving - b cutting - c filing - d bending - e abrading - f vacuum forming - g deforming and reforming.
	4.7.3 Fabricating/constructing/assembling: - a tapping/threading - b fastening – use of nuts, bolts and washers - c use of adhesives – contact adhesive, epoxy resin, Tensol® cement, liquid cement (dichloromethane) - d wastage - e addition.